SERVER

```
import java.io.*;
import java.net.*;

public class VirtualServer {
    public static void main(String[] args) {
        int port = 9000;

        try (ServerSocket serverSocket = new ServerSocket(port)) {
            System.out.println("VirtualServer running on port " + port);

            while (true) {
                Socket clientSocket = serverSocket.accept();

                new Thread(() -> handleClient(clientSocket)).start();
            }

        } catch (IOException e) {
            e.printStackTrace();
        }
    }

    private static void handleClient(Socket socket) {
        try {
            BufferedReader in = new BufferedReader(new InputStreamReader(socket.getInputStream()));
            OutputStream out = socket.getOutputStream();



            String requestLine = in.readLine();
            if (requestLine == null) return;
```

```
String hostHeader = "";
String line;
while (!(line = in.readLine()).isEmpty()) {
    if (line.toLowerCase().startsWith("host:")) {
        hostHeader = line.substring(5).trim();
    }
}


String responseBody;
switch (hostHeader) {
    case "server1.com":
        responseBody = "Response from VIRTUAL SERVER 1";
        break;
    case "server2.com":
        responseBody = "Response from VIRTUAL SERVER 2";
        break;
    default:
        responseBody = "Response from DEFAULT SERVER";
}


String response = "HTTP/1.0 200 OK\r\n" +
                "Content-Type: text/plain\r\n" +
                "\r\n" +
                responseBody;

out.write(response.getBytes());
out.flush();
```

```
            socket.close();

        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```

PROXY

```
import java.io.*;
import java.net.*;
import java.util.*;

public class MultiServerProxy {

    public static void main(String[] args) {
        int proxyPort = 8000;

        try (ServerSocket proxySocket = new ServerSocket(proxyPort)) {
            System.out.println("MultiServerProxy running on port " + proxyPort);

            while (true) {
                Socket clientSocket = proxySocket.accept();

                new Thread(() -> handleClient(clientSocket)).start();
            }

        } catch (IOException e) {
            e.printStackTrace();
        }
```

```
    }

    private static void handleClient(Socket clientSocket) {
        try {
            BufferedReader clientIn = new BufferedReader(new
InputStreamReader(clientSocket.getInputStream()));
            OutputStream clientOut = clientSocket.getOutputStream();



            String requestLine = clientIn.readLine();
            if (requestLine == null) return;

            Map<String, String> headers = new HashMap<>();
            String line;
            while (!(line = clientIn.readLine()).isEmpty()) {
                int idx = line.indexOf(":");
                if (idx > 0) {
                    headers.put(line.substring(0, idx).trim(), line.substring(idx + 1).trim());
                }
            }



            String hostHeader = headers.get("Host");
            String serverHost = "localhost";
            int serverPort = 9000;

            System.out.println("Proxy: forwarding request for " + hostHeader);



            Socket serverSocket = new Socket(serverHost, serverPort);
            OutputStream serverOut = serverSocket.getOutputStream();
```

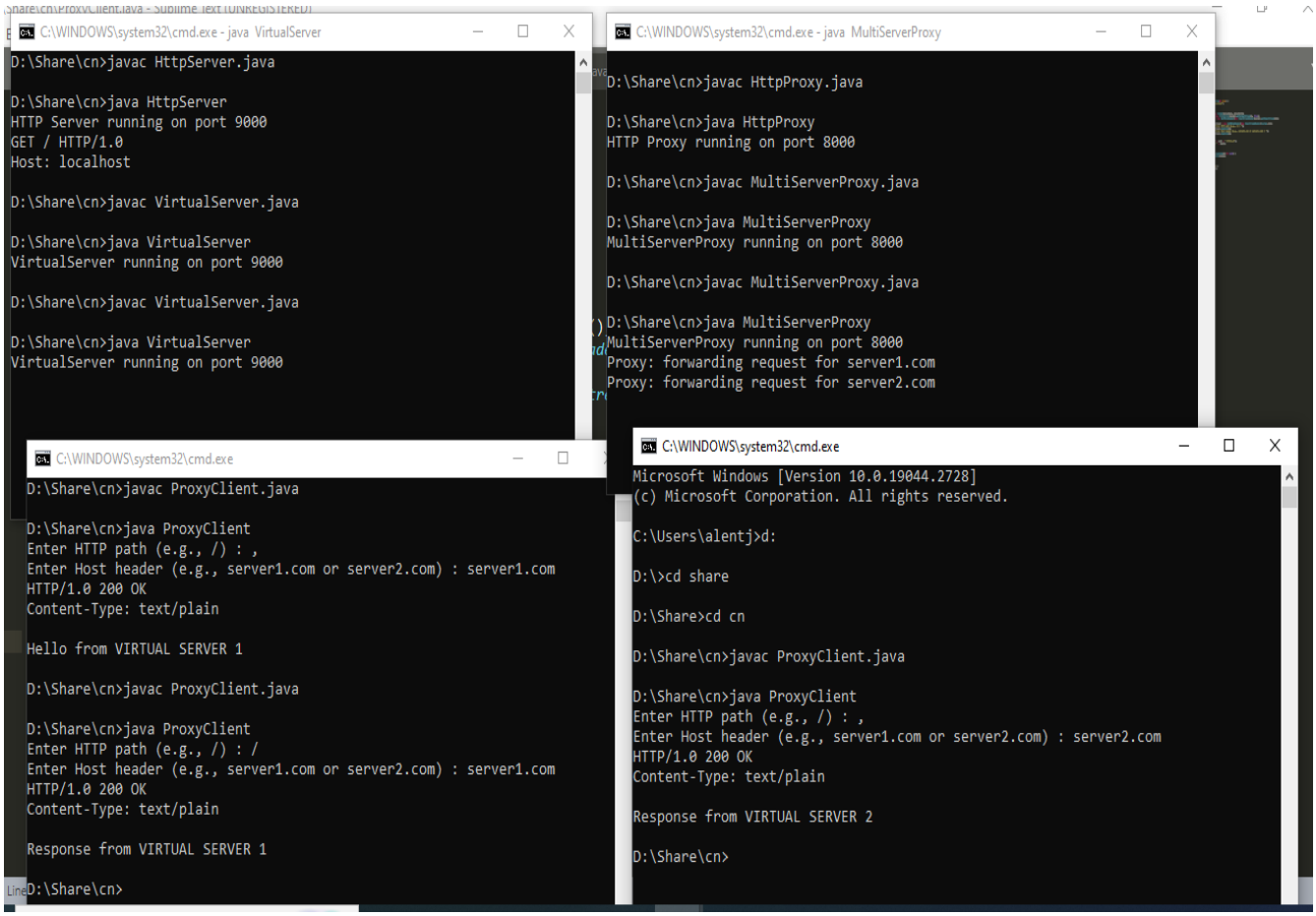
```
            InputStream serverIn = serverSocket.getInputStream();



            serverOut.write((requestLine + "\r\n").getBytes());
            for (Map.Entry<String, String> entry : headers.entrySet()) {
                serverOut.write((entry.getKey() + ": " + entry.getValue() + "\r\n").getBytes());
            }
            serverOut.write("\r\n".getBytes());
            serverOut.flush();



            byte[] buffer = new byte[4096];
            int bytesRead;
            while ((bytesRead = serverIn.read(buffer)) != -1) {
                clientOut.write(buffer, 0, bytesRead);
            }

            clientOut.flush();
            clientSocket.close();
            serverSocket.close();

        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```

CLIENT

```
import java.io.*;
import java.net.*;
```

```
public class ProxyClient {
    public static void main(String[] args) {
        String proxyHost = "localhost";
        int proxyPort = 8000;

        try {
            Socket socket = new Socket(proxyHost, proxyPort);
            PrintWriter out = new PrintWriter(socket.getOutputStream(), true);
            BufferedReader in = new BufferedReader(new InputStreamReader(socket.getInputStream()));

            BufferedReader userInput = new BufferedReader(new InputStreamReader(System.in));
            System.out.print("Enter HTTP path (e.g., /) : ");
            String path = userInput.readLine();
            System.out.print("Enter Host header (e.g., server1.com or server2.com) : ");
            String host = userInput.readLine();



            out.println("GET " + path + " HTTP/1.0");
            out.println("Host: " + host);
            out.println();

            String line;
            while ((line = in.readLine()) != null) {
                System.out.println(line);
            }

            socket.close();
        } catch (IOException e) {
            e.printStackTrace();
        }
```

```
    }
}
```


C:\WINDOWS\system32\cmd.exe - java VirtualServer
D:\Share\cn>javac HttpServer.java
D:\Share\cn>java HttpServer
HTTP Server running on port 9000
GET / HTTP/1.0
Host: localhost
D:\Share\cn>javac VirtualServer.java
D:\Share\cn>java VirtualServer
VirtualServer running on port 9000
D:\Share\cn>javac VirtualServer.java
D:\Share\cn>java VirtualServer
VirtualServer running on port 9000
C:\WINDOWS\system32\cmd.exe - java MultiServerProxy
D:\Share\cn>javac HttpProxy.java
D:\Share\cn>java HttpProxy
HTTP Proxy running on port 8000
D:\Share\cn>javac MultiServerProxy.java
D:\Share\cn>java MultiServerProxy
MultiServerProxy running on port 8000
D:\Share\cn>javac MultiServerProxy.java
D:\Share\cn>java MultiServerProxy
MultiServerProxy running on port 8000
Proxy: forwarding request for server1.com
Proxy: forwarding request for server2.com
C:\WINDOWS\system32\cmd.exe
D:\Share\cn>javac ProxyClient.java
D:\Share\cn>java ProxyClient
Enter HTTP path (e.g., /) : ,
Enter Host header (e.g., server1.com or server2.com) : server1.com
HTTP/1.0 200 OK
Content-Type: text/plain
Hello from VIRTUAL SERVER 1
D:\Share\cn>javac ProxyClient.java
D:\Share\cn>java ProxyClient
Enter HTTP path (e.g., /) : /
Enter Host header (e.g., server1.com or server2.com) : server1.com
HTTP/1.0 200 OK
Content-Type: text/plain
Response from VIRTUAL SERVER 1
D:\Share\cn>
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 10.0.19044.2728]
(c) Microsoft Corporation. All rights reserved.
C:\Users\alentj>d:
D:\>cd share
D:\Share>cd cn
D:\Share\cn>javac ProxyClient.java
D:\Share\cn>java ProxyClient
Enter HTTP path (e.g., /) : ,
Enter Host header (e.g., server1.com or server2.com) : server2.com
HTTP/1.0 200 OK
Content-Type: text/plain
Response from VIRTUAL SERVER 2
D:\Share\cn>